

LOGBOOK

Entry 1: April 14, 2026 – Problem Identification

Today I started brainstorming for the Science Expo. I noticed that everyone at school uses the same basic passwords, usually just their name and a "1!". I want to build something that proves why this is dangerous. I'm thinking of a tool that doesn't just check for "rules" (like capital letters) but actually measures how hard a password is to guess.

- **Goal:** Create a Computer Science project that helps people make better passwords.

Entry 2: April 21, 2026 – Research Phase

I spent time researching how hackers actually break passwords. I found out about "Brute Force" and "Dictionary Attacks." I also discovered a library called `zxcvbn` (created by Dropbox) that uses math to guess how many "attempts" a computer would need to crack a password. This is much better than basic rules.

- **New Concept learned:** Entropy (the measure of randomness). I need to figure out how to turn "guesses" into "bits of entropy."

Entry 3: April 28, 2026 – The Privacy Problem

I wanted to add a feature that checks if a password has been leaked in a real data breach (like the LinkedIn or Facebook leaks). But then I realized: **How can I check a user's password without stealing it?** I found a solution online: The **Have I Been Pwned API**. It uses something called "k-Anonymity." I will only send the first 5 characters of a password's hash. This feels like a major "Engineering" win for my project.

Entry 4: May 4, 2026 – First Prototype (Drafting the UI)

I started coding the HTML and CSS today. I want a "Cybersecurity" feel—dark background and neon colors. I hit a snag: the "meter bar" wasn't moving when I typed. I realized I forgot to link the JavaScript to the HTML input. Fixed it!

- **Sketching:** (Draw a quick rectangle here with an input box and a bar underneath).

Entry 5: May 5, 2026 – Integrating the API

Today was tough. I had to learn how to use the Web Crypto API to generate a SHA-1 hash. I got an error: Uncaught (in promise). It turns out I didn't wait for the API to respond before trying to show the results. I added `async` and `await` to my functions, and now it works!

- **Observation:** When I type "password", it instantly says "BREACH DETECTED". It's working!

Entry 6: May 6, 2026 – Final Refinements & Ethics

I added a "Password Visibility" toggle (the eye icon) because it's a standard security feature. I also filled out my **Learner Ethics Checklist**. Since I'm not saving any passwords or using real people's names, my project is "Low Risk."

I'm calling the final prototype "**SentinelVault**."